Revealing Breast Cancer Causing Key-Genes through Survival Analysis for Diagnosis and Therapies

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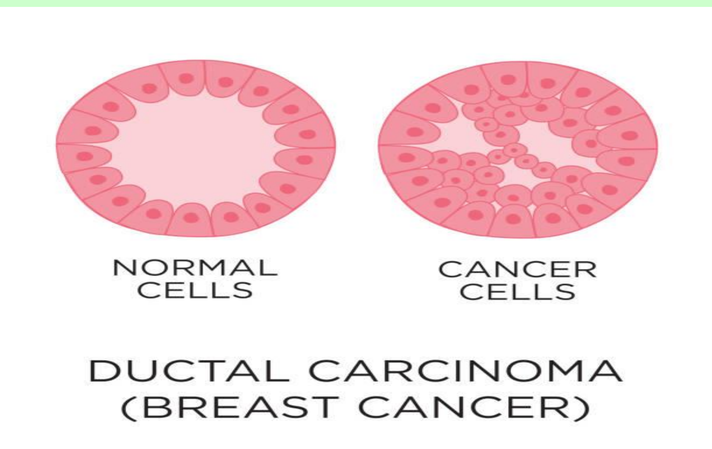
Introduction

Breast cancer is one of the most prevalent cancers worldwide, and following its treatment, many women turn to plastic surgery for reconstruction. A diagnosis of cancer is a heavy burden on patients. Our aim to revealing breast cancer causing Key-genes(KGs) through survival analysis for diagnosis and therapies. we collected breast cancer causing KGs suggested in different published articles by the literature review. We found a total of 83 independent articles that suggested breast cancer causing KGs-set. We collected 83 KGs-sets from the selected 83 articles. We found a total of 314 unique KGs from those KGs-sets and used them for further investigation in this study. Then we verify their differential expression patterns between breast cancer and control samples by investigating the survival probability curves and box plots with the TCGA database and found 41 DEGs. Then we identified top-ranked key DEGs and transcription factors(TF) ,miRNA and finally suggest the top effective drugs of the disease.

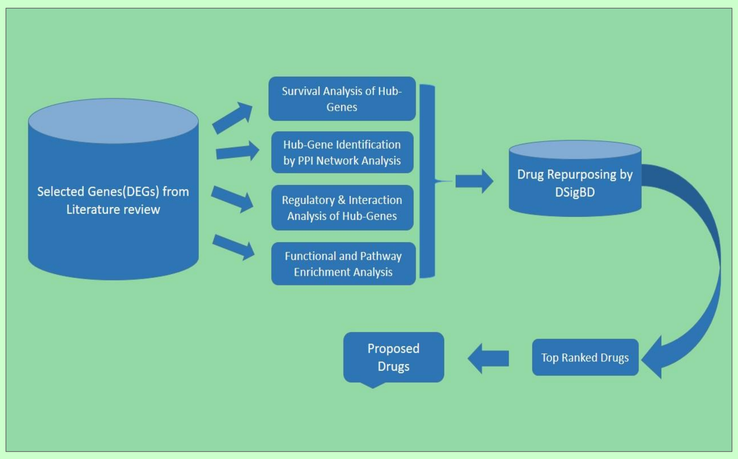


Objectives of this Study

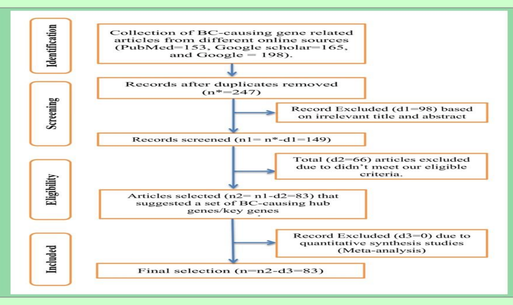
The objective of a survey is one of the most important things. Therefore, objectives of my research are given below:

- **Exploring Breast Cancer Causing Top-Ordered Core Genes**
- **Literature Review for Diagnosis and Therapies**

Pipeline of this Study



Pipeline of Data collection



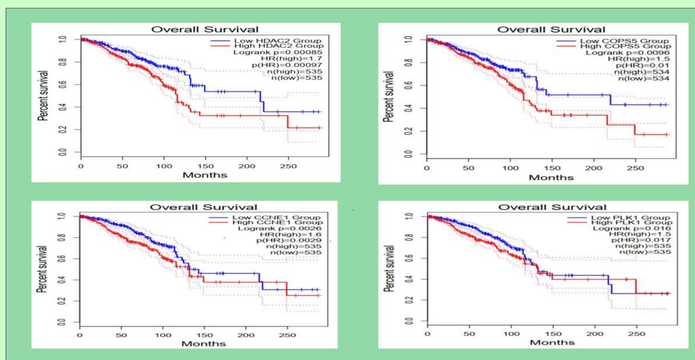
Methods

- Survival Analysis
- Gene Ontology and Pathway Enrichment Analysis
- Protein-Protein Interaction and Key Gene Identification
- TF-gene and Gene-miRNA Interactions
- Candidate Drug Screening by DSigBD

Result

Survival Analysis

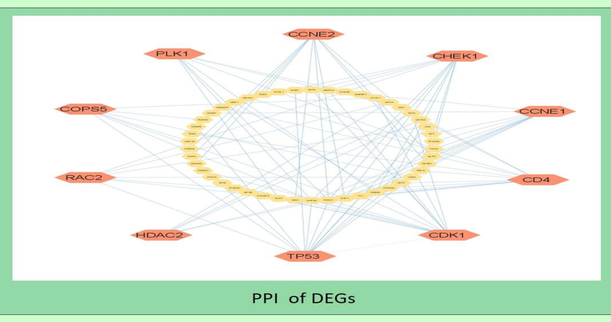
After this analysis 41 DEGs are founded which are significantly expressed and their p-value <0.05.



Survival analysis

Protein-Protein Interaction (PPI) Network Analysis

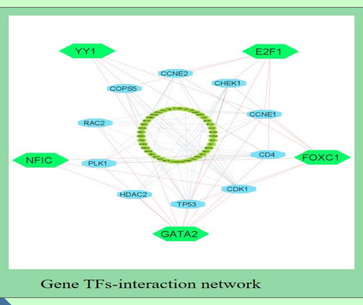
40 DEGs were found after putting 41 DEGs in STRING and **10** Key Genes **(TP53, COPS5 , CCNE2 , CCNE1 , PLK1 , HDAC2 , CD4 , CDK1 , RAC2 , CHEK1)** are found that resulate all other DEGs.



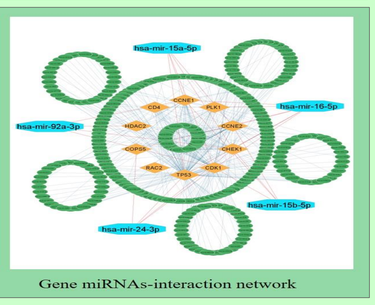
PPI of DEGs

TF-Gene and Gene-miRNA Network Analysis

Transcription Factor-gene interaction network was built around protein encoded by hub proteins from NetworkAnalyst. JASPAR from NetworkAnalyst was used to compile TF-gene interactions. The gene regulatory network analysis revealed **5 miRNAs** (hsa-mir-16-5p, hsa-mir-15a-5p, hsa-mir-24-3p, hsa-mir-92a-3p, hsa-mir-15b-5p) and **5 TFs** (GATA2, FOXC1, E2F1, YY1, NFIC) as the transcriptional and post-transcriptional regulators of KGs**.**



Gene TFs-interaction network



Gene miRNAs-interaction network

Gene ontology (GO) and KEGG Pathway Analysis

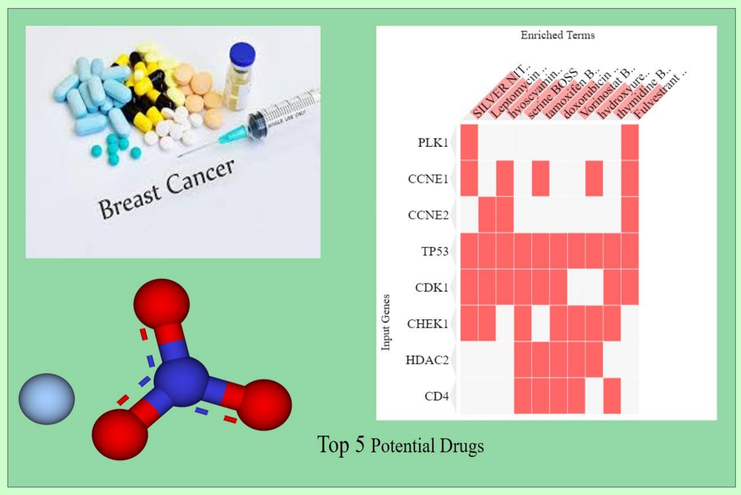
We used the Enrichr online tool to perform GO and KEGG pathway analysis. Functional enrichment analyses were performing to explain the biological processes, cellular components molecular functions and KEGG pathways related to proteins determined by the DEGs.

Category	Term	PValue	Associated key Genes
GOTERM_BP_DIRECT	GO:0043066~negative regulation of apoptotic process	5.64E-05	HDAC2, COPS5, PLK1, CDK1, TP53
GOTERM_BP_DIRECT	GO:0006468~protein phosphorylation	3.53E-04	CCNE1, PLK1, CHEK1, CDK1
GOTERM_CC_DIRECT	GO:0005813~centrosome	7.91E-04	CD4,TP53
GOTERM_CC_DIRECT	GO:0005654~nucleoplasm	5.22E-06	CCNE2, CCNE1, PLK1, CHEK1, CDK1, TP53
GOTERM_MF_DIRECT	GO:0019901~protein kinase binding	1.41E-04	HDAC2, PLK1, COPS5, CCNE2, CCNE1, CHEK1, CDK1, RAC2, TP53
GOTERM_MF_DIRECT	GO:0005515~protein binding	3.52E-07	CD4, CCNE2, CCNE1, PLK1, RAC2, TP53

Pathway Name	PValue	Associated key Genes
hsa05166:Human T-cell leukemia virus 1 infection	3.61E-07	CD4,CCNE2, CCNE1, CHEK1,TP53
hsa04110:Cell cycle	2.77E-06	HDAC2, CCNE2, CCNE1, PLK1, CHEK1, CDK1, TP53

Drug Repurposing

- Based on p-value ,the 6 potential drug molecules for the treatment against breast cancer are **SILVER NITRATE CTD,Leptomycin B CTD, hyoscyamine CTD , serine BOSS, tamoxifen BOSS, doxorubicin BOSS**.



Top 5 Potential Drugs

Conclusion

We investigated the top causing core genes for breast cancer diagnosis and therapies. The result we found are correlated with other studies.By this investigation we found some key biomarkers that are mainly responsible for breast cancer. We also research about the diagnosis of the responsible genes that are influenced theproblem. The suggested drugs prognosis cancer by halting the regulation of expressing genes.

Reference

1. Hong, Z.; Wang, Q.; Hong, C.; Liu, M.; Qiu, P.; Lin, R.; Lin, X.; Chen, F.; Li, Q.; Liu, L.; Wang, C.; Chen, D., Identification of Seven Cell Cycle-Related Genes with Unfavorable Prognosis and Construction of their TF-miRNA-mRNA regulatory network in Breast Cancer. J Cancer 2021, 12, (3). doi: 10.7150/jca.48245, 740-753.
2. Zeng, X.; Shi, G.; He, Q.; Zhu, P., Screening and predicted value of potential biomarkers for breast cancer using bioinformatics analysis. Sci Rep 2021, 11, (1). doi: 10.1038/s41598-021-00268-9, 20799.

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